Exercise 6 :

6.1) by direct calculation,

$$g(\vec{\omega}) = \| \vec{y} - \dot{X}\vec{\omega} \|^2 = \vec{y}^T\vec{y} - 2\vec{y}^T\dot{X}\vec{\omega} + (\dot{X}\vec{\omega})^T\dot{X}\vec{\omega}$$

$$= \vec{\omega}^T\dot{X}^T\dot{X}\vec{\omega} - 2\vec{y}\dot{X}\vec{\omega} + \vec{y}^T y$$

$$= \frac{1}{2}\vec{\omega}^T(2\dot{X}^T X)\vec{\omega} - 2\vec{y}\dot{X}\vec{\omega} + \vec{y}^T\vec{y}$$

$A = 2\dot{X}^T X$ $\qquad$ $g(\vec{\omega}) = \frac{1}{2}\vec{\omega}^T A\vec{\omega} - 2\vec{y}\dot{X}\vec{\omega} + \vec{y}^T\vec{y}$

6.2) We first compute ∇g :

$$\nabla g(\vec{\omega}) = \frac{2\times 1}{2} A\vec{\omega} - 2\vec{y}^T\dot{X} + 0$$

$$= A\vec{\omega} - 2\vec{y}^T\dot{X}$$

so by direct computation, we have :

$$H(\vec{\omega}) = \nabla^2 g(\vec{\omega}) = A + 0$$

and then $\nabla^2 g(\vec{\omega}) = A$

• $\forall \vec{z}$, $\vec{z}^T A z = \vec{z}^T(2\dot{X}^T X)\vec{z} = 2\vec{z}^T X^T X\vec{z} = 2\|X^T\vec{z}\|^2 \geq 0$

$\vec{z}^T A\vec{z} \geq 0$ hence A is positive semi definite.